

Tech Stack

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Overview

LoveYou is a full-stack web application following a client-server architecture. The system consists of a React frontend, an Express backend, and a PostgreSQL database.

Frontend

Technology	Purpose
React	Build reusable user interfaces
Vite	Fast frontend development and build tool
TypeScript	Static type checking
Axios	HTTP communication with backend APIs
React Router DOM	Client-side routing

Responsibility

The frontend provides the graphical user interface for users, including:

- Authentication
 - Profile management
 - Matching
 - Messaging
 - Other dating features
-

Backend

Technology	Purpose
Node.js	JavaScript runtime
Express.js	REST API framework

Responsibility

The backend handles:

- Business logic
- Authentication
- Authorization
- User management
- API endpoints
- Database access

Database

Technology	Purpose
PostgreSQL	Relational database
Prisma ORM	Database modeling and querying

Responsibility

Store:

- Users
- Profiles
- Matches
- Messages
- Other application data

Authentication

Technology	Purpose
JWT	User authentication
bcrypt	Password hashing

Passwords are securely hashed before storage.

Email Service

Technology	Purpose
Nodemailer	Sending emails

Used for email-related features such as account verification or notifications.

Validation

Technology	Purpose
Zod	Request validation

Ensures incoming API requests satisfy predefined schemas.

Version Control

Technology	Purpose
Git	Version control
GitHub	Source code hosting and collaboration

Architecture Style

The project follows a typical three-tier architecture.

Presentation Layer

- React

Business Layer

- Express API

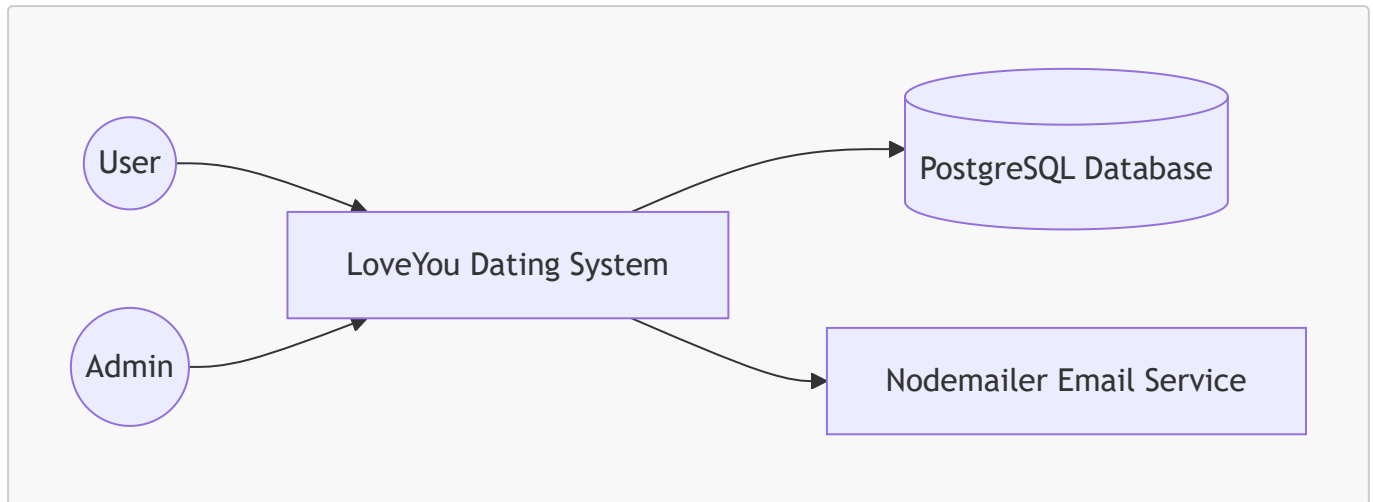
Data Layer

- PostgreSQL + Prisma

C4 Model Level 1 - System Context Diagram

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System Context Diagram



Description

LoveYou Dating System

LoveYou is a dating web application that allows users to register, authenticate, manage profiles, find matches, and communicate with other users.

User

Users interact with the system through the web interface.

Main activities include:

- Register
 - Login
 - Update profile
 - Browse profiles
 - Send messages
 - Manage account
-

Admin

Administrators manage the platform.

Responsibilities include:

- Monitor users
 - Manage system data
 - Perform administrative operations
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PostgreSQL Database

The database stores all persistent information, including:

- User accounts
 - Profile information
 - Messages
 - Matching information
-

Email Service

The backend communicates with Nodemailer to send emails for notification-related features.

Interaction Summary

- Users access the system through the web interface.
- The backend processes requests and accesses PostgreSQL through Prisma.
- Email notifications are sent through Nodemailer.